Vectors in n Dimensions

Vector addition and scalar multiplication

1 Let $\mathbf{u} = \langle 2, -1, 3 \rangle$, $\mathbf{v} = \langle 1, 4, -2 \rangle$. Find $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.

2 Let $\mathbf{a} = \langle 1, 0, -2, 3 \rangle$, $\mathbf{b} = \langle -1, 2, 1, 0 \rangle$ (vectors in $\mathbb{R}^4$). Find $\mathbf{a} + 2\mathbf{b}$.

3 Solve for $\mathbf{x}$: $\mathbf{x} + \langle 2, -3 \rangle = \langle 5, 1 \rangle$.

Dot product and norm

4 For $\mathbf{u} = \langle 1, 2, 2 \rangle$, $\mathbf{v} = \langle 3, -1, 0 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$ and $|\mathbf{u}|$.

5 Find the angle between $\mathbf{u} = \langle 1, 0, 0 \rangle$ and $\mathbf{v} = \langle 1, 1, 0 \rangle$ using $\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

6 For $\mathbf{u} = \langle 2, 2, -1 \rangle$, $\mathbf{v} = \langle 0, 3, 4 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$, $|\mathbf{u}|$, and $|\mathbf{v}|$.

The Cauchy–Schwarz Inequality

7 State the Cauchy–Schwarz Inequality.

8 Verify the Cauchy–Schwarz Inequality for $\mathbf{u} = \langle 3, 4 \rangle$, $\mathbf{v} = \langle 1, 2 \rangle$.

9 For which pairs $\mathbf{u}, \mathbf{v}$ does Cauchy–Schwarz hold with EQUALITY? Verify equality for $\mathbf{u} = \langle 2, 4 \rangle$, $\mathbf{v} = \langle 1, 2 \rangle$.

Unit vectors

10 Find the unit vector in the direction of $\mathbf{v} = \langle 3, -4 \rangle$.

11 Find a unit vector in the direction of $\mathbf{v} = \langle 1, 2, 2 \rangle$.

12 Find a vector of length 10 in the same direction as $\mathbf{v} = \langle 3, 4 \rangle$.

Orthogonality

13 Determine whether $\mathbf{u} = \langle 2, -3, 1 \rangle$ and $\mathbf{v} = \langle 1, 2, 4 \rangle$ are orthogonal.

14 For $\mathbf{u} = \langle 1, -1, 2 \rangle$, find $|\mathbf{u}|$ and the unit vector $\hat{\mathbf{u}}$.

15 Find a value of k so that $\mathbf{u} = \langle k, 3 \rangle$ is orthogonal to $\mathbf{v} = \langle 4, -2 \rangle$.

The triangle inequality and vector geometry

16 State the Triangle Inequality for vectors.

17 Verify the Triangle Inequality for $\mathbf{u} = \langle 3, 0 \rangle$, $\mathbf{v} = \langle 0, 4 \rangle$.

Projections

18 The projection of $\mathbf{u}$ onto $\mathbf{v}$ is $\mathrm{proj}_{\mathbf{v}}\mathbf{u} = \dfrac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}\mathbf{v}$. Find $\mathrm{proj}_{\mathbf{v}}\mathbf{u}$ for $\mathbf{u} = \langle 3, 4 \rangle$, $\mathbf{v} = \langle 1, 0 \rangle$.

19 Find $\mathrm{proj}_{\mathbf{v}}\mathbf{u}$ for $\mathbf{u} = \langle 4, 3 \rangle$, $\mathbf{v} = \langle 1, 1 \rangle$.

Linear combinations and span

20 Write $\mathbf{w} = \langle 7, 4 \rangle$ as a linear combination $a\mathbf{u} + b\mathbf{v}$ of $\mathbf{u} = \langle 1, 0 \rangle$ and $\mathbf{v} = \langle 0, 1 \rangle$.

21 Write $\mathbf{w} = \langle 5, 1 \rangle$ as a linear combination $a\mathbf{u} + b\mathbf{v}$ of $\mathbf{u} = \langle 1, 1 \rangle$ and $\mathbf{v} = \langle 1, -1 \rangle$.

22 Explain briefly why every vector in $\mathbb{R}^2$ can be written as a linear combination of $\mathbf{u} = \langle 1, 1 \rangle$ and $\mathbf{v} = \langle 1, -1 \rangle$ (i.e., why they span $\mathbb{R}^2$).

23 Is $\mathbf{w} = \langle 3, 6 \rangle$ in the span of $\mathbf{u} = \langle 1, 2 \rangle$ alone? Justify your answer.

24 Explain why $\mathbf{u} = \langle 1, 2 \rangle$ and $\mathbf{v} = \langle 2, 4 \rangle$ do NOT span all of $\mathbb{R}^2$.

25 Determine whether $\mathbf{w} = \langle 1, 1, 1 \rangle$ is a linear combination of $\mathbf{u} = \langle 1, 0, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 0 \rangle$.

26 Write $\mathbf{w} = \langle 4, -1, 7 \rangle$ as a linear combination of $\mathbf{u} = \langle 1, 0, 0 \rangle$, $\mathbf{v} = \langle 0, 1, 0 \rangle$, and $\mathbf{z} = \langle 0, 0, 1 \rangle$, and explain why this is always possible for ANY vector in $\mathbb{R}^3$.

27 How many vectors, at minimum, are needed to span $\mathbb{R}^n$? Justify briefly.

28 Determine whether $\mathbf{u} = \langle 1, 2, -1 \rangle$, $\mathbf{v} = \langle 2, 1, 1 \rangle$, and $\mathbf{w} = \langle 3, 3, 0 \rangle$ span $\mathbb{R}^3$, given that $\mathbf{w} = \mathbf{u} + \mathbf{v}$.

29 Give an example of three vectors in $\mathbb{R}^3$ that DO span $\mathbb{R}^3$, and briefly justify why they work.

30 Summarize in one sentence the difference between a set of vectors being linearly independent and a set of vectors spanning a space.